Reversible Solid Oxide Cells as a Mars Power Solution

Human missions to Mars require massive amounts of power, oxygen, and propellant, all of which must be produced on Earth, as each kilogram launched from Earth comes with a high logistical cost. This paper looks at the Reversible Solid Oxide Fuel Cell (RSOFC) as a full approach to that challenge. The RSOFC operates at 600-900°C on a yttria-stabilized zirconia electrolyte. It switches between electrolysis mode, which splits Martian CO_2 into oxygen and carbon monoxide during times of surplus power. When in fuel cell mode, which uses stored fuel to generate electricity during dust storms and at night. Combined with the Sabatier reaction that makes methane from the Martian CO_2 and water ice. This system produces energy, oxygen, rocket fuel, and water from a single device. The waste heat at 800°C can also offset habitat heating. The overall system efficiency can reach close to 81%, including the habitat heating. From 2021 until 2023, NASA's MOXIE experiment confirmed the main technology aboard Perseverance, producing 122 grams of 98% pure oxygen over 16 runs, more than doubling the initial target. The suggested scale-up, "Big Atmospheric MOXIE," aims at 3 kg per hour. The RSOFC does not solve just one Mars problem. It solves nearly every one of them.

Mars Environment Challenges

Mars presents extreme environmental constraints. Its atmosphere is composed of around 95% carbon dioxide, with a surface pressure less than 1% of Earth's, as shown in Fig. 1. The temperature averages around -60°C. Dust storms could remain for months and even shut down solar-powered systems completely. NASA's Opportunity rover stopped communicating following the 2018 global dust storm, as prolonged dust accumulation reduced solar power below operational thresholds. For any future human presence on the surface, these are not minor inconveniences. There are significant engineering challenges that must be solved before a single astronaut sets foot on the ground.

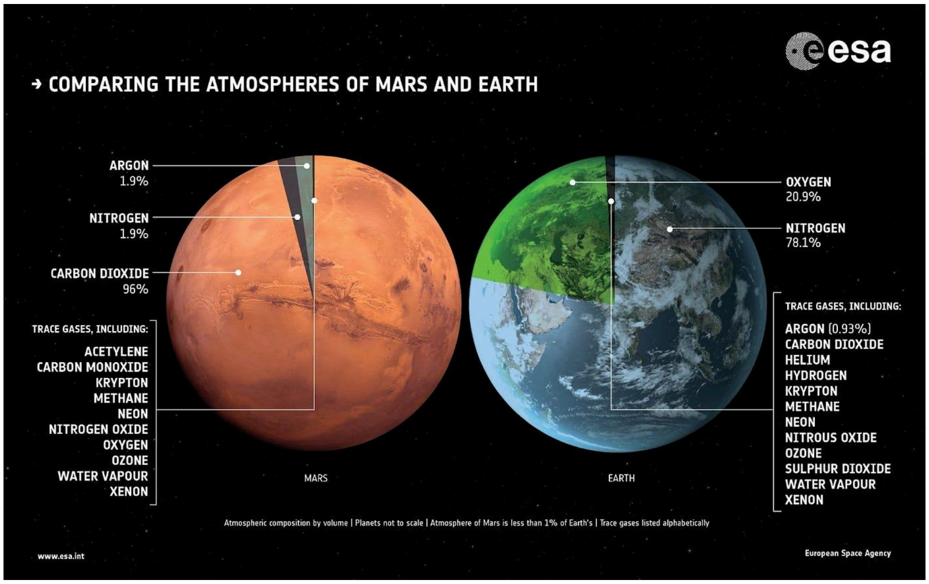

Fig.1

Power Requirements and Challenges

Power is the first issue. Assuming a crewed Mars base of four to six people would need about 21 to 25 kilowatts of continuous electric power to power the habitat, life support systems, and research equipment (Reddy, 2023; Hoffman et al., 2023). While solar may be capable of doing this on clear days, Mars is not always cooperative with the weather. When a dust storm hits the planet, which might last for weeks, solar production can drop by up to 99 percent. Batteries could be the solution, but the math quickly becomes problematic. Hundreds of metric tons of battery mass can be needed to store enough energy to support an entire crew during a month-long storm (Guan et al., 2006). Launching even a single kilogram to the Martian surface requires placing roughly 12-13 kilograms into low Earth orbit first. Batteries at that scale are simply not a viable option (Hoffman et al., 2022).

Limitations of Standard Fuel Cells

Standard fuel cells do not address the issue either. One of the most well-known forms of fuel cells is the proton exchange membrane (PEM) fuel cell. This type of Fuel cell runs on pure hydrogen and generates energy through an oxygen reaction. Mars lacks abundant hydrogen, and its high CO_2 levels make PEM operation nearly impossible. While hydrogen can be easily produced from water ice formations, using it solely to power a standard fuel cell wastes a valuable resource. That locally sourced hydrogen is far better used in combination with Martian CO_2 to produce methane and O_2, the exact propellants that SpaceX's Raptor engines run on. Every standard fuel cell either misses Mars's full potential or fails to withstand the Martian environment. This requires a systems-level power architecture that utilizes its existing resources while also producing something useful.

Reversible Solid Oxide Fuel Cells (RSOFC)

What Mars needs is a reversible solid oxide fuel cell. The RSOFC extends solid oxide fuel cell technology by enabling reversible operation at high temperatures. While a typical (PEM) fuel cell requires pure hydrogen and operates at roughly 80 degrees Celsius, it is highly sensitive to carbon monoxide contamination. This can significantly degrade performance. The Reversible Solid Oxide Fuel Cell (RSOFC) works great under the conditions found on Mars. It operates at temperatures ranging from 600 to 900 degrees Celsius. The RSOFC runs solely on carbon dioxide and carbon monoxide, and can produce methane, oxygen, and water, depending on its use (Sabbaghi et al., 2025). NASA's MOXIE experiment traveled aboard the Perseverance rover and showed solid oxide electrolysis of Martian CO_2 between 2021 and 2023. The experiment produced oxygen 16 times at purities above 98 percent, and rates double NASA's original targets. MOXIE was a proof of concept (NASA Jet Propulsion Laboratory, 2023). What comes next is a scaled system capable of supporting human life and filling the propellant tanks of the vehicles that bring crews home.

RSOFC Chemistry and Electrolyte Function

Solid oxide fuel cells use a completely different approach. The electrolyte is a dense ceramic, most often yttria-stabilized zirconia (YSZ). At room temperature, YSZ offers nothing beneficial. Heating it to 600-900 degrees Celsius opens the crystal structure, allowing oxide ions (O^{2-}) to migrate from one electrode to the other, which is necessary for both electricity generation and fuel production on Mars

At the cathode, oxygen receives electrons from the external circuit and forms O^{2-}. This step allows CO_2 from the Martian atmosphere to be split, producing oxygen when running in electrolysis mode. Those ions move through the YSZ to the anode, where they react with the available fuel, either carbon monoxide or methane, depending on the chosen reaction mode. Running on carbon monoxide: $CO + O^{2-} \rightarrow CO_2 + 2e^-$ or running on methane: $CH_4 + 4O^{2-} \rightarrow CO_2 + 2H_2O + 8e^-$. Produces electricity and water, which could support life or be further split into hydrogen and oxygen. Both reactions release electrons as electricity (Han et al., 2007).

The primary advantage comes directly from this chemistry. Because the anode runs at 800 degrees Celsius, it can oxidize almost any carbon-based fuel. Carbon monoxide, which would rapidly destroy a PEM cell, is not a contaminant but a fuel. This specific aspect suggests the possibility of using the Martian atmosphere as a power source. The system simultaneously produces O_2 or methane when needed

The high operating temperature also eliminates the need for expensive platinum catalysts. This enables internal reforming of hydrocarbon fuels and generates high-quality waste heat for combined heat and power systems. The trade-off is that ceramic components must withstand severe temperatures without degrading, making thermal management an important technical challenge.

RSOFC Advantages and Reversibility

In addition, the RSOFC is reversible, meaning the same device that generates energy from fuel can also produce and store fuel for future use. In an environment where dust storms can reduce solar power generation to almost nothing for months at a time, the ability to store and release energy on demand is a necessity. In a typical fuel cell, current flows in one direction: fuel in, electricity out. In a reversible solid oxide cell, the direction can be changed. In a fuel cell mode, it consumes fuel and generates power. In Solid Oxide Electrolysis Cell (SOEC) mode, it consumes power while producing fuel. Both modes use the same ceramic electrolyte, electrodes, and operating temperature (Petri, 2012).

During excess generation on sunny days and a nuclear reactor's output, the RSOFC runs in electrolyzer mode. Electricity and CO_2 enter the cell, and the output fuel is

stored. When a dust storm hits and solar power declines, the system changes to fuel cell mode. Fuels that have been stored return through the same device, while electricity is released. One device performs both roles, reducing the overall cost of hauling separate electrolyzer and fuel cell equipment. This is essential on a mission when each kilogram launched from Earth must first place 12 to 13 kilograms in low Earth orbit (Hoffman et al., 2022). DOE-funded testing confirmed this in actual hardware. A daily cycling test lasted over 11,900 hours, replicating a solar-integrated system cycling 10.5 hours in electrolyzer mode and 12.5 hours in fuel cell mode per day over 6,000 mode-switching cycles, or more than 15 years of daily operation (Petri, 2012).

MOXIE: Proof of Concept on Mars

When NASA's Perseverance rover landed in early 2021, it was carrying a small but important payload. The Mars Oxygen In-Situ Resource Utilization Experiment (MOXIE). MOXIE worked on the Martian surface from April 2021 to September 2023, contributing to a new leap in space exploration. It was not just a piece of equipment, it was the first successful example of "living off the land" on another planet. By the end of the mission, MOXIE had not only met its goals but also almost doubled the output NASA expected (NASA).

MOXIE collected the thin Martian atmosphere through a HEPA filter, compressed it, and heated it to 800°C before feeding it into the Solid Oxide Electrolysis (SOXE) assembly. Carbon dioxide was stripped over a nickel-ceramic cathode in this unit, producing oxygen ions and carbon monoxide. Those oxygen ions were then conducted through a scandia-stabilized zirconia (ScSZ) electrolyte to the anode, where they recombined and were released as pure oxygen. ScSZ operates on the same principle as YSZ dopant oxides, which create vacancies in the crystal lattice that allow oxide ions to migrate, but scandia's ionic radius more closely matches zirconia's, producing higher ionic conductivity at the same operating temperature. The results across sixteen different runs were outstanding. MOXIE produced 122 grams of oxygen in total, with a peak production rate of 12 grams per hour, roughly double the original goal. Above all, it proved its capacity to withstand the extreme swings of Martian temperatures, changing air densities, and the transition from day to night (NASA Jet Propulsion Laboratory, 2023).

From Prototype to "BAM"

What mattered most about the mission was not just how much oxygen MOXIE produced, but what happened to the system over time. As the studies continued, researchers noticed a steady increase in electrical resistance within the electrolysis stack. They eventually linked it down to thermal cycling. MOXIE could not keep heated due to the rover's low power supply. Instead, it had to be heated to around 800°C for each run, then cooled again. That continual heating and cooling served as a stress test that a full-scale system on a human mission would not face. These units would run continuously, avoiding the repeated "heat shock" that caused most of the degradation detected on the rover. Looking ahead, the "Big Atmospheric MOXIE," or BAM, comes into the picture. The concept is straightforward, use what worked on a small scale and make it durable enough to support astronauts. BAM is intended to produce approximately 3 kilograms of oxygen per hour, an important step toward meeting mission demands. This is not a small device, weighing over 9,000 kilograms and consuming nearly 27 kilowatts of electricity. However, when you consider the alternative, the data begin to make a lot more sense. A Mars Ascent Module requires approximately 31 tons of oxygen solely as propellant, and bringing it from Earth would require launching 500 tons into low Earth orbit, which is almost impossible. Producing oxygen directly on Mars completely changes the equation. This not only makes missions more efficient but also more realistic.

The Sabatier Reaction

Only half of the propellant equation involves oxygen. The Raptor engines on the Starship run on liquid methane and liquid oxygen, commonly known as methalox, so methane must be produced on Mars, too. This requires one step above ordinary electrolysis, the Sabatier reaction. At roughly 300°C, CO_2 and hydrogen react with a nickel catalyst to produce methane and water ($CO_2 + 4H_2 \rightarrow CH_4 + 2H_2O$). It is exothermic, and the water it creates can be electrolyzed back into hydrogen and oxygen, allowing the system to function partially self-sufficiently Hoffman et al. (2022).

The hydrogen source comes from Martian water ice. Subsurface permafrost has been proven at the poles and observed across a wide range of latitudes. Extraction and electrolysis of that water using surplus electrical power provides hydrogen for the reactor, while the oxygen produced is used for both life support and propellant storage. Methane was chosen as the Starship's propellant because it can be synthesized from Martian resources. The objective has always been to land the vehicle mostly empty, generate fuel locally, and launch without returning propellant from Earth.

Power Management

The primary energy sources for a Mars base are deployable solar panels and small fission reactors. Solar alone cannot be trusted. Dust storms can shut down production for weeks, which is where fission comes in. Reactors like NASA's Kilopower project use a uranium core to supply consistent, weather-independent power regardless of what the planet's sky looks like. The two sources work together to keep the RSOFC operating. During surplus generation, the process electrolyzes Martian CO_2 into methane and oxygen. When a storm hits and solar power fails, it shifts to fuel cell mode and draws electricity from stored propellant. A crew of six need just around 0.2 kilos of oxygen per hour to breathe, which is a fraction of the 3 kilograms generated in a full-scale system, with the remainder flowing directly into propellant tanks (NASA, 2023).

The RSOFC operates at 800°C, and a significant percentage of the energy passing through it is dissipated as heat instead of electricity. On a planet with an average surface temperature of -60°C, this is not a waste but a benefit. Heat captured via an exchanger can be directed through the habitat's thermal management system, which minimizes or eliminates the requirement for specific electric heaters while preheating the Sabatier reactor inputs. A standalone solid oxide fuel cell attains an electrical efficiency of 39 to 54 percent; with waste heat recovery, the total system can achieve an efficiency exceeding 81 percent. On a planet where every kilowatt must be created locally or launched from Earth, this difference significantly changes how the entire power system is sized.

Engineering Challenges

Several challenges exist, but none are considered major obstacles. Maintaining ceramic components at 800°C in a -60°C environment needs sophisticated insulation. Heating and cooling cycles cause mechanical stress on the YSZ ceramic, needing more long-term testing under Martian-like conditions. Chromium evaporation from metallic interconnects poisons cathode activity steadily over time, and those connections need to operate for more than 80,000 hours, an amount that has yet to be tested at mission size. Martian dust, which is fine and electrostatically charged, requires filter systems that can operate for years without maintenance. Atmospheric pressure of about 600 pascals also causes significant gas compression, which accounts for roughly 40 percent of MOXIE's power draw.

Conclusion

MOXIE proved the concept with 16 runs, 122 grams of oxygen, 98% purity, and twice NASA's original objective. A full-scale system that works continuously would avoid heat cycling, which is the only source of true damage. The engineering hurdles are serious but achievable. Looking ahead, future development will focus on increasing ceramic lifespans beyond 80,000 hours, upgrading connector materials, and directly connecting the RSOFC to fission power to eliminate solar dependence totally. On the environmental side, every kilogram of oxygen and methane produced on Mars means one less kilogram launched from Earth, thereby lowering both mission costs and the carbon footprint of space exploration. The RSOFC produces no toxic byproducts, consumes CO_2 that blankets the entire globe, and provides only what humans require for survival. It transforms Mars' atmosphere from a danger to a supply chain.

References

Guan, J., et al. (2006). *High performance flexible reversible solid oxide fuel cell* (Final Technical Report No. DE-FC36-04GO14351). U.S. Department of Energy Office of Scientific and Technical Information. https://www.osti.gov/servlets/purl/899650

Han, M., Tang, X., Yin, H., & Peng, S. (2007). Fabrication, microstructure and properties of a YSZ electrolyte for SOFCs. *Journal of Power Sources, 165*(2), 757–763. https://doi.org/10.1016/j.jpowsour.2006.11.054

Hoffman, J. A., Hecht, M. H., Rapp, D., Hartvigsen, J. J., SooHoo, J. G., Aboobaker, A. M., McClean, J. B., Liu, A. M., Hinterman, E. D., Nasr, M., Hariharan, S., Horn, K. J., Meyen, F. E., Okkels, H., Steen, P., Elangovan, S., Graves, C. R., Khopkar, P., Madsen, M. B., . . . Eisenman, D. J. (2022). Mars Oxygen ISRU Experiment (MOXIE) Preparing for human Mars exploration. *Science Advances, 8*(35), eabp8636. https://doi.org/10.1126/sciadv.abp8636

Hoffman, J. A., Hinterman, E. R., Hecht, M. H., Rapp, D., & Hartvigsen, J. J. (2023). 18 months of MOXIE (Mars oxygen ISRU experiment) operations on the surface of Mars Preparing for human Mars exploration. *Acta Astronautica, 210*, 547–553. https://doi.org/10.1016/j.actaastro.2023.04.045

NASA. (2023). *Kilopower*. NASA Space Technology Mission Directorate. https://www.nasa.gov/directorates/stmd/tech-demo-missions-program/kilopower-hmqzw/

NASA Jet Propulsion Laboratory. (2023, September 6). *NASA's oxygen-generating experiment MOXIE completes Mars mission*. https://www.jpl.nasa.gov/news/nasas-oxygen-generating-experiment-moxie-completes-mars-mission

Petri, R. (2012). *Advanced materials for reversible solid oxide fuel cell (RSOFC), dual-mode operation with low degradation* (DOE Hydrogen and Fuel Cells Program FY 2012 Annual Progress Report, V.J.3). Versa Power Systems. U.S. Department of Energy. https://www.hydrogen.energy.gov

Reddy, M. J. (2023). *Metallic materials for solid oxide fuel cells and electrolysers: Mitigating high temperature corrosion* [Doctoral thesis, Chalmers University of Technology]. Chalmers Research. ISBN 978-91-7905-832-6.

Sabbaghi, M. A., Genceli, H., & Baniasadi, E. (2025). Evaluating the economic viability and technological feasibility of a new SOFC-integrated energy system using CO_2 capture unit. *International Journal of Hydrogen Energy, 144*, 674-685. https://doi.org/10.1016/j.ijhydene.2025.02.360